

Lab 11: A One-Dimensional GAN Sketch

Two Gaussians as data. Watch a tiny generator miss a mode.

Work in a **Jupyter** notebook with **PyTorch**. Do notes **11.1–11.3** first. CPU is enough; keep the nets tiny and stop around a few hundred steps.

First time you implement this. The lecture notes this week already answered what the method is, why it exists, the architecture, the formula, and the tradeoffs. Read that first-time block before these exercises. This lab is the first time you **compute** it, not the first definition.

```
import torch
from torch import nn
import matplotlib.pyplot as plt

torch.manual_seed(0)

MEANS = torch.tensor([[-2.0, 0.0], [2.0, 0.0]])

def real_batch(n=64):
    m = torch.randint(0, 2, (n,))
    return MEANS[m] + 0.25 * torch.randn(n, 2)
```

Two well-separated 2-D Gaussians. That is the whole dataset. You are checking whether G covers **both** modes, not whether it paints faces.

1. Tiny MLP G and D

```
G = nn.Sequential(nn.Linear(8, 32), nn.ReLU(), nn.Linear(32, 2))
D = nn.Sequential(nn.Linear(2, 32), nn.ReLU(), nn.Linear(32, 1))
optG = torch.optim.Adam(G.parameters(), lr=1e-3)
optD = torch.optim.Adam(D.parameters(), lr=1e-3)
bce = nn.BCEWithLogitsLoss()
ones = lambda n: torch.ones(n, 1)
zeros = lambda n: torch.zeros(n, 1)
```

1. Confirm `real_batch(512)` really has two clouds: scatter-plot it. Mark the means.
2. Draw `z = torch.randn(512, 8)`, plot `G(z)` **before** training. It should look like a blob at the origin, not like the data.

2. A short adversarial loop

For 400 steps (you may stop at 200 if the plot already moves):

```
n = 64
for step in range(400):
    real = real_batch(n)
    fake = G(torch.randn(n, 8)).detach()
    lossD = bce(D(real), ones(n)) + bce(D(fake), zeros(n))
    optD.zero_grad(); lossD.backward(); optD.step()

    fake = G(torch.randn(n, 8))
    lossG = bce(D(fake), ones(n))
    optG.zero_grad(); lossG.backward(); optG.step()
```

Print `lossD` and `lossG` every 100 steps. Do not treat them as quality.

After training, scatter 512 reals and 512 fakes on one plot (two colors).

3. Did you miss a mode?

Assign each fake to the nearest of the two means. Report the fraction in each bin.

1. If either bin is under 10% of the fakes, write **collapse** (or heavy imbalance). If both sit near 50%, write **covered**.
2. Train once more from a new seed, or with 800 steps, only if the first run is a pancake on one mean. One extra run is enough. This lab is allowed to collapse; reporting it is the point.
3. Four sentences: what G and D each wanted, why a low generator loss is not coverage, which mode (if any) you missed, and one change you would try (noise dim, D steps, or more training) without claiming it is a cure.